Machine Learning

Binary Classification

Python · sklearn · SHAP

Streamlit Deployment

WATER QUALITY CLASSIFICATION

Safe Drinking Water Prediction

Predicting potability from 9 physicochemical water parameters

PROBLEM STATEMENT

The Challenge

Access to safe drinking water is a critical global health issue. Traditional water testing methods are:

- Time-consuming and costly
- Require specialized lab equipment
- Not scalable to remote areas

Machine learning offers a faster, scalable alternative to classify water as potable or non-potable.

Project Objective

Build a binary classifier that predicts whether water is safe to drink based on 9 measurable physicochemical parameters.

Key goals:

- Compare 4 ML algorithms
- Handle missing data & class imbalance
- Explain model decisions via SHAP
- Deploy as an interactive web app

DATASET OVERVIEW

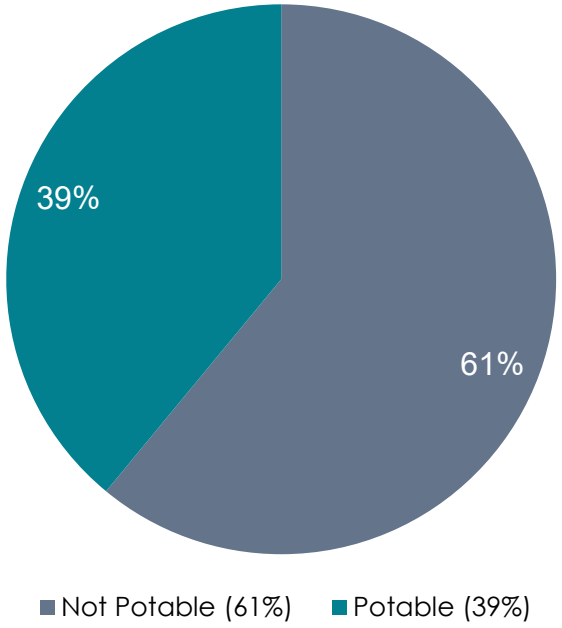
3,276	9	39%	3
Total Samples	Features	Potable (Class 1)	Cols w/ Missing

Dataset Features (water_potability.csv)

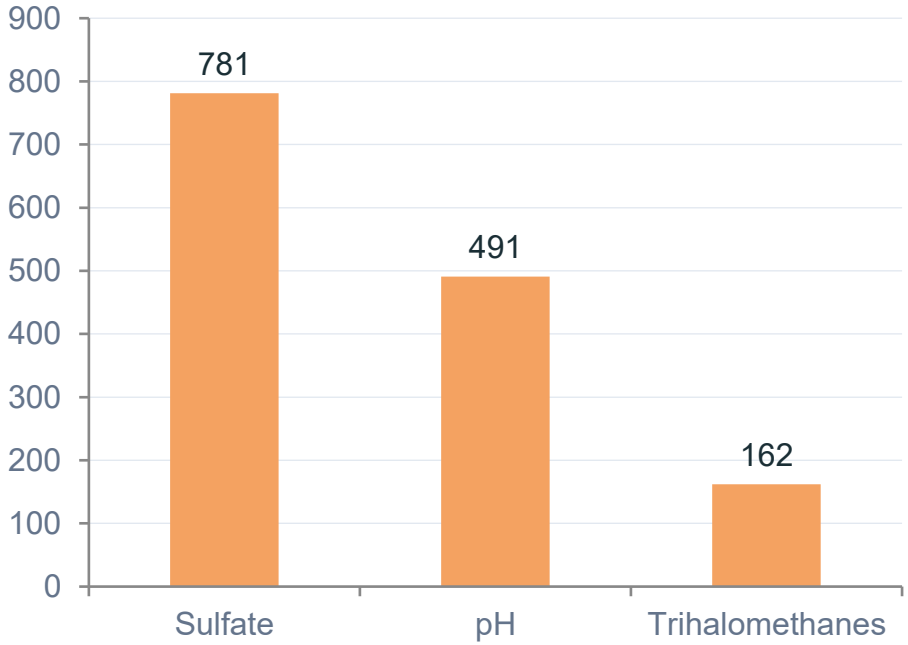
Feature	Description	Nulls?
pH	Acidity / alkalinity of water (0–14)	**491**
Hardness	Calcium & magnesium mineral content	0
Solids	Total dissolved solids (TDS) in ppm	0
Chloramines	Chloramine disinfectant amount	0
Sulfate	Sulfate ion concentration	**781**
Conductivity	Electrical conductivity in µS/cm	0
Organic Carbon	Total organic carbon in ppm	0
Trihalomethanes	Disinfection byproduct concentration	**162**
Turbidity	Light scattering measure (NTU)	0

EXPLORATORY DATA ANALYSIS

Class Distribution



Missing Value Counts



Key Insight: Dataset has significant class imbalance (61% non-potable vs 39% potable) → addressed with SMOTE oversampling

DATA PREPROCESSING PIPELINE

01 KNN Imputation

Missing values in pH (491), Sulfate (781), and Trihalomethanes (162) filled using K-Nearest Neighbours (k=5) — preserves local data structure vs. mean imputation.

02 Robust Scaling

Features scaled using RobustScaler — uses median & IQR, making it resistant to outliers in water sensor data where extreme readings are common.

03 Train/Test Split

80/20 stratified split ensures class proportions are preserved in both training (2,620 samples) and test sets (656 samples).

04 SMOTE Oversampling

Applied on training data only: Synthetic Minority Oversampling creates synthetic potable samples → balances to 1,598 vs 1,598 per class.

After SMOTE: Training set → 3,196 samples | Class 0: 1,598 · Class 1: 1,598 (perfectly balanced)

MODELS EVALUATED

Logistic Regression

Baseline

Linear model estimating probability of potability using a sigmoid function. Fast, interpretable, weak on non-linear patterns.

solver=liblinear, random_state=42

Random Forest

Best Model ★

Ensemble of decision trees using bagging. Captures non-linear feature interactions. Robust to outliers and noise.

n_estimators=100 (default), random_state=42

Gradient Boosting

Ensemble

Boosting ensemble — builds trees sequentially, each correcting previous errors. Strong learner from weak base models.

n_estimators=100, learning_rate=0.1 (default)

SVM

Kernel Method

Support Vector Machine finding the optimal hyperplane margin. Effective in high-dimensional spaces. Slowest to train.

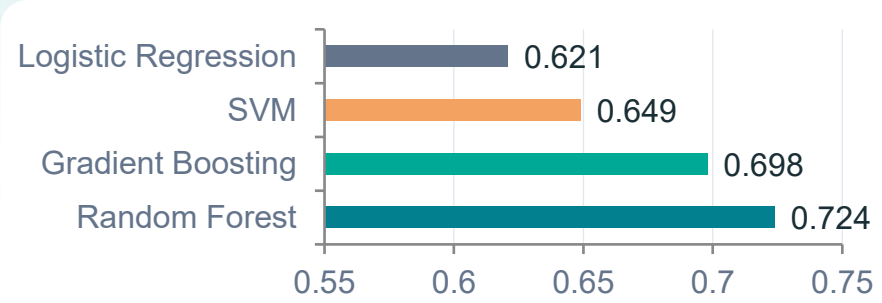
kernel=rbf (default), probability=True

MODEL PERFORMANCE RESULTS

Evaluation on Hold-out Test Set (656 samples)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest ★	0.6800	0.6012	0.5820	0.5914	**0.7240**
Gradient Boosting	0.6585	0.5761	0.5664	0.5712	0.6980
Logistic Regression	0.5945	0.4783	0.4961	0.4870	0.6210
SVM	0.6174	0.5093	0.5352	0.5220	0.6487

ROC-AUC Comparison



Accuracy Comparison

SHAP INTERPRETABILITY

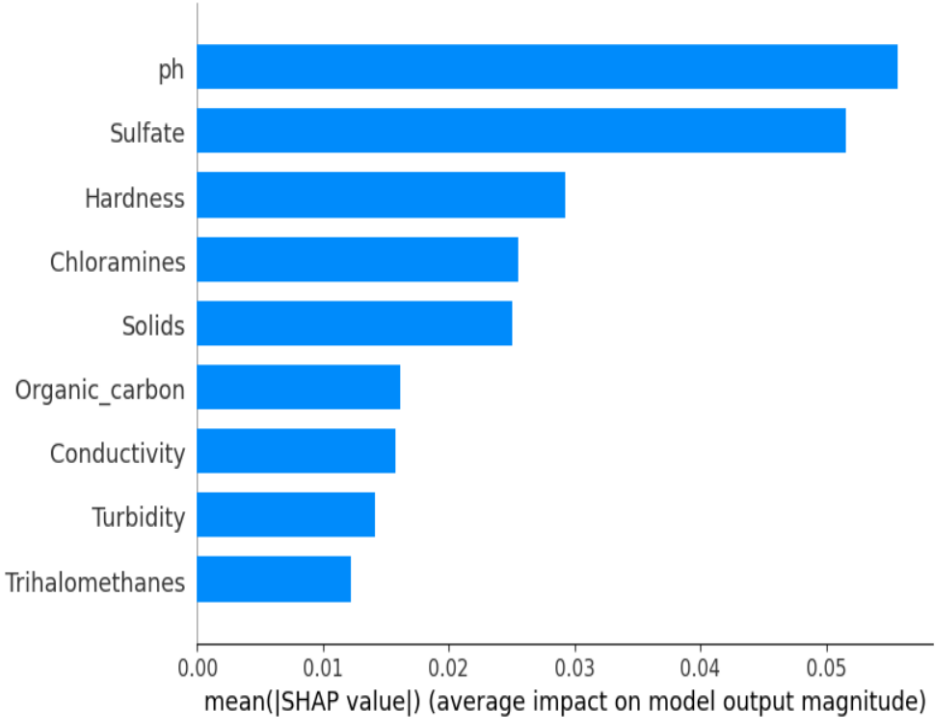
What is SHAP?

SHAP (SHapley Additive exPlanations) assigns each feature a contribution value for a specific prediction.

Based on cooperative game theory:
- TreeExplainer used for Random Forest
- Provides both global & local explanations
- Reveals which features drive potability

Why it matters:
ML models for public health decisions must be explainable and auditable — not black boxes.

Feature Importance (Mean |SHAP| values)



Sulfate, pH, and Solids are the most influential features in determining water potability.

DEPLOYMENT

Live Inference Pipeline

Data Input → Impute Nulls → Robust Scale → RF Predict → Output

rf_model.pkl

Serialized Random Forest classifier trained on balanced SMOTE data

scaler.pkl

Fitted RobustScaler — ensures new inputs are scaled identically to training data

app.py

Streamlit web application — user inputs 9 water parameters, returns potability prediction

requirements.txt

All Python dependencies stored for reproducible deployment environment

github.com/evans-0/water_quality_classification — Clone, install requirements.txt, run **streamlit run app.py**

KEY TAKEAWAYS

- ✓ Random Forest achieved best performance — ROC-AUC 0.724, Accuracy 68%
- ✓ KNN imputation + SMOTE effectively handled missing data and class imbalance
- ✓ SHAP revealed Sulfate, pH & Solids as top predictive features
- ✓ Model deployed as an interactive Streamlit web application

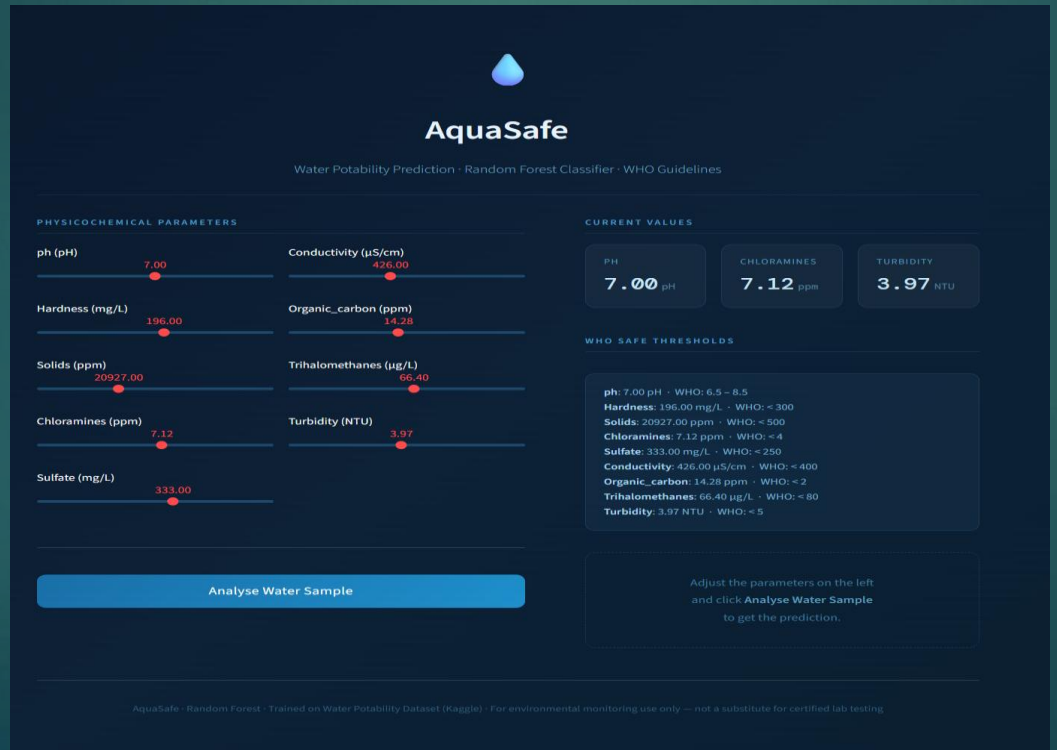
Future Work

Hyperparameter tuning (GridSearchCV) · Deep learning models · Real-time sensor API integration · Expand to multi-class water quality levels

Water Quality Classification

3,276
samples analyzed

72.4%
Best ROC-AUC

Live Demo

- Streamlit Deployment
  https://waterqualityclassification-project1.streamlit.app/